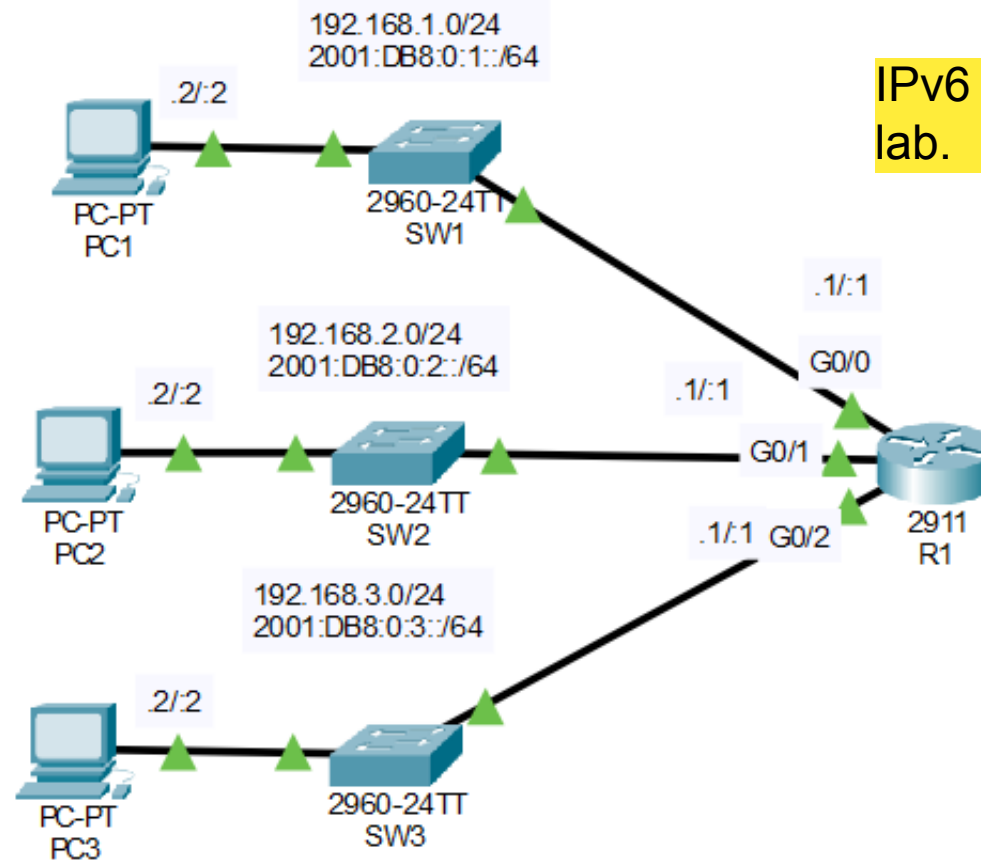




IPv6 network topology used for the lab.

The IPv4 configuration of each device is complete. Perform the following IPv6 configurations to create an IPv4/IPv6 'dual-stack' network.

1. Enable IPv6 routing on R1.
2. Configure the appropriate IPv6 addresses on R1.
3. Confirm your configurations.
What IPv6 addresses are present on each interface?
4. Configure the appropriate IPv6 addresses on each PC.
Configure the correct default gateway.
5. Attempt to ping between the PCs (IPv4 and IPv6)

Task 2 & 3 Complete — Configured IPv6 addresses on R1's interfaces.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0
R1(config-if)#ipv6 address 2001:db8:0:1::1/64
R1(config-if)#int g0/1
R1(config-if)#ipv6 2001:db8:0:2::1/64
                        ^
% Invalid input detected at '^' marker.

R1(config-if)#ipv6 address 2001:db8:0:2::1/64
R1(config-if)#int g0/2
R1(config-if)#ipv6 address 2001:db8:0:3::1/64
R1(config-if)#
R1(config-if)#|
```

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0030.A3B0.AC87

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.3.2

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address 2001:DB8:0:3::2 / 64

Link Local Address: FE80::230:A3FF:FE80:AC87

Task 4 Complete — Configured the IPv6 address on PC3.

Cisco Packet Tracer PC Command Line 1.0

C:\>

C:\>

C:\>ping 2001:db8:0:1::1

TGask 5 Complete — Successfully tested IPv6 connectivity with 0% packet loss.

Pinging 2001:db8:0:1::1 with 32 bytes of data:

Reply from 2001:DB8:0:1::1: bytes=32 time<1ms TTL=255

Reply from 2001:DB8:0:1::1: bytes=32 time<1ms TTL=255

Reply from 2001:DB8:0:1::1: bytes=32 time<1ms TTL=255

Reply from 2001:DB8:0:1::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:0:1::1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:db8:0:2::1

Pinging 2001:db8:0:2::1 with 32 bytes of data:

Reply from 2001:DB8:0:2::1: bytes=32 time<1ms TTL=255

Reply from 2001:DB8:0:2::1: bytes=32 time=4ms TTL=255

Reply from 2001:DB8:0:2::1: bytes=32 time<1ms TTL=255

Reply from 2001:DB8:0:2::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:0:2::1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>

Enabled IPv6 unicast routing on R1.

```
R1 (config)#ipv6 unicast-routing
```

Successfully pinged PC2 using IPv6 with 0% packet loss.

```
C:\>ping 2001:db8:0:2::2
```

```
Pinging 2001:db8:0:2::2 with 32 bytes of data:
```

```
Reply from 2001:DB8:0:2::2: bytes=32 time<1ms TTL=127
```

```
Reply from 2001:DB8:0:2::2: bytes=32 time<1ms TTL=127
```

```
Reply from 2001:DB8:0:2::2: bytes=32 time<1ms TTL=127
```

```
Reply from 2001:DB8:0:2::2: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 2001:DB8:0:2::2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>
```